School of Natural Sciences and Mathematics

The Biology Program at UT Dallas emphasizes the unifying molecular and cellular nature of organisms. At the center of the Biology undergraduate curriculum are the biochemical, genetic, and cell biology concepts and tools used to study the genes of prokaryotes and eukaryotes, to study the proteins and ribonucleic acids (RNA) encoded by these genes, and to study how the expression of these genes is regulated during the development and lifetimes of organisms. Molecular Biology represents a fusion of the four disciplines of biochemistry, biophysics, genetics, and cell biology. Modern biology requires a background in other disciplines such as chemistry, mathematics, physics, and computer sciences. Principles from these disciplines have to be merged to understand and apply new biotechnology and genetic engineering techniques. It is desirable for entering students to have a broad interest and background in the sciences.

Molecular Biology (BS)

A BS degree is offered in Molecular Biology. The BS degrees are intended as preparation for scientific careers in biology or careers in the health professions. Biology offers a streamlined double major with Business Administration or Criminology. Accelerated BS/MS Fast Track programs are available that provide the opportunity to partially satisfy the requirements of a master's degree while completing the bachelor's degree in Molecular Biology.

Bachelor of Science in Molecular Biology

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

Faculty

FACG> nsm-molecular-biology-bs

Professors: Juan E. González, Kelli Palmer, Lawrence J. Reitzer, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Joseph Boll, Mehmet Candas, Nikki Delk, Tae Hoon Kim, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicole De Nisco, Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Erica Sanchez, Darshan Sapkota

Professors Emeriti: Lee A. Bulla, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison

Professors of Instruction: Scott Rippel, Uma Srikanth

Associate Professors of Instruction: Wen Lin, Elizabeth Pickett, Ilya Sapozhnikov

Assistant Professors of Instruction: Stephanie Boyd, Yi Huang, Meenakshi Maitra, Caitlin Maynard, Iti Mehta, Ramesh Padmanabhan, Jing Pan, Ruben Ramirez, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu

Associate Professor of Practice: Uyen Henson

Research Assistant Professors: Li Liu, Ru-Hung Wang

Senior Lecturer: Wen-Ho Yu

UT Dallas Affiliated Faculty: Heng Du, Jung-whan (Jay) Kim

I. Core Curriculum Requirements: 42 semester credit hours[1]

Communication: 6 semester credit hours

Select any 6 semester credit hours from Communication Core courses (see advisor)

Mathematics: 3 semester credit hours

MATH 2417 Calculus I[2, 3, 4]

Or select any 3 semester credit hours from Mathematics Core courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

CHEM 1311 General Chemistry I[2]

or CHEM 1315 Honors Freshman Chemistry I[2]

CHEM 1312 General Chemistry II[2]

or CHEM 1316 Honors Freshman Chemistry II[2]

Or select any 6 semester credit hours from Life and Physical Sciences Core courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from Language, Philosophy and Culture Core courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from Creative Arts Core courses (see advisor)

American History: 6 semester credit hours

Select any 3 semester credit hours from American History Core courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from Government/Political Science Core courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from Social and Behavioral Sciences Core courses (see advisor)

Component Area Option: 6 semester credit hours

BIOL 2311 Introduction to Modern Biology I[2]

MATH 2419 Calculus II[2, 3, 4]

Or select any 6 semester credit hours from Component Area Option Core courses (see advisor)

II. Major Requirements: 68-70 semester credit hours

Major Preparatory Courses: 24-25 semester credit hours beyond Core Curriculum

CHEM 1111 General Chemistry Laboratory I

or CHEM 1115 Honors Freshman Chemistry Laboratory I

CHEM 1112 General Chemistry Laboratory II

or CHEM 1116 Honors Freshman Chemistry Laboratory II

CHEM 1311 General Chemistry I[2]

or CHEM 1315 Honors Freshman Chemistry I[2]

CHEM 1312 General Chemistry II[2]

or CHEM 1316 Honors Freshman Chemistry II[2]

CHEM 2323 Introductory Organic Chemistry I[5] or CHEM 2327 Honors Organic Chemistry I[5]

CHEM 2325 Introductory Organic Chemistry II[5]

or CHEM 2328 Honors Organic Chemistry II[5]

CHEM 2233 Introductory Organic Chemistry Laboratory[5]

or CHEM 2237 Honors Organic Chemistry Laboratory[5]

MATH 2417 Calculus I[2, 3, 4]

MATH 2419 Calculus II[2, 3, 4]

MATH 2418 Linear Algebra

PHYS 2325 Mechanics

and PHYS 2125 Physics Laboratory I

or PHYS 2421 Honors Physics I - Mechanics and Heat[6]

PHYS 2326 Electromagnetism and Waves

or PHYS 2422 Honors Physics II - Electromagnetism and Waves

PHYS 2126 Physics Laboratory II

Major Core Courses: 32-33 semester credit hours beyond Core Curriculum

BIOL 2111 Introduction to Modern Biology Workshop I[5]

BIOL 2112 Introduction to Modern Biology Workshop II[5]

BIOL 2281 Introductory Biology Laboratory[5]

BIOL 2311 Introduction to Modern Biology I[2, 5]

BIOL 2312 Introduction to Modern Biology II[5]

BIOL 3401 Genetics

BIOL 3402 Molecular and Cell Biology

BIOL 3461 Biochemistry I

or CHEM 3461 Biochemistry I

BIOL 3462 Biochemistry II

or CHEM 3462 Biochemistry II

or BIOL 3335 Microbial Physiology

BIOL 3380 Biochemistry Laboratory

BIOL 4380 Cell and Molecular Biology Laboratory[7]

or BIOL 3V96 Undergraduate Research in Molecular and Cell Biology[7] (3 semester credit hours)

or BIOL 4399 Senior Honors Research for Thesis in Molecular and Cell Biology[7]

or BIOL 4391 Senior Research in Molecular and Cell Biology[7]

BIOL 4461 Biophysical Chemistry

Major Related Courses: 12 semester credit hours[8]

12 semester credit hours upper-division approved molecular biology-related BIOL or CHEM electives

III. Elective Requirements: 8-10 semester credit hours

Free Electives: 8-10 semester credit hours

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. The available Fast Track options and their admission requirements, if any, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- MS in Molecular and Cell Biology: Students must have completed at least 15 semester credit hours of upper-division Biology core courses.
- MS in Biotechnology: Students must have a cumulative GPA of at least 3.00 in all courses.

- MS in Bioinformatics and Computational Biology: Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- MS in Data Science and Statistics (Only Data Science and Applied Statistics specializations): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

Interested students, after reviewing the UT Dallas Fast Track policy, should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the Fast Track program.

Degree Planning

Upper-division biology courses taken at other institutions may be included as part of the degree plan subject to the provisions of the section on Transfer Admissions.

Major-related courses may not include more than 9 semester credit hours (BS) or 6 semester credit hours (BA) of upper-division transfer credit and not more than 3 semester credit hours (Biology major) or 6 semester credit hours (Molecular Biology major) of individual instruction (e.g., BIOL 3V90, BIOL 3V91, BIOL 3V96, BIOL 4302, BIOL 4390, BIOL 4391, BIOL 4399, or BIOL 4V99).

Students planning a career in a particular allied health profession should consult the school they expect to attend to apprise themselves of the course requirements for admission.

Admission standards for medical and dental schools are set by the individual professional school, whose specific requirements should be reviewed with the help of the UT Dallas Health Professions Advising Center (HPAC). Most professional schools prefer that admission applications be channeled through the HPAC.

1. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

2. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

3. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option and 2 semester credit hours of Calculus are counted as Major Preparatory Courses.

4. Students may choose one of the following calculus sequences: (a) MATH 2413, MATH 2414, and MATH 2415; or (b) MATH 2417 and MATH 2419.

5. Indicates a prerequisite class to be completed before enrolling for upper-division classes.

6. Students who complete PHYS 2421 do not need to complete PHYS 2125.

7. These substitutes for BIOL 4380 require permission of the Biology Undergraduate Faculty

Advisor to ensure equivalent training in recombinant DNA analysis.

8. Up to 6 semester credit hours of research may be used in fulfilling the major related course requirement.

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